

## project Report

**Title:** Graph Neural Networks for Metabolic Pathway Analysis

**Course:** Bioinformatics (2025)

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### 1. Introduction

The objective of this project is to implement a Graph Neural Network (GCN) to predict enzyme participation in reactions within metabolic pathways of *Saccharomyces cerevisiae*. Each enzyme (gene) is represented as a node, biological relationships between enzymes are represented as edges, and the task is to classify enzymes into two categories:

- Label 0: Enzyme participates in only one reaction
- Label 1: Enzyme participates in more than one reaction

This project provides hands-on experience with:

- Constructing biological graphs from KEGG pathway data
- Implementing GNN layers from scratch
- Training and evaluating graph-based models

The biological motivation is that enzymes act in concert within pathways, and predicting multi-reaction participation can highlight biologically important enzymes.

### 2. Data Preprocessing

#### 2.1 Data Acquisition

- Organism: *Saccharomyces cerevisiae*
- Data Source: KEGG (Kyoto Encyclopedia of Genes and Genomes)
- KGML Files Downloaded:
- sce00010, sce00020, sce00030, sce00620, sce01200, sce01230

These KGML files encode enzyme nodes, reactions, and enzyme–enzyme relationships. Only gene entries (enzymes) were used as nodes; compounds and other entries were ignored.

## 2.2 Graph Construction

- Nodes: 222 enzymes
- Edges: 1180 directed (590 unique undirected)
- Isolated Nodes: 98 nodes (cannot propagate messages)
- Connected Components: 99 components; largest has 124 nodes (~56%)
- Edge Density: 0.024 (sparse graph)

## 2.3 Feature Engineering

Each node is represented by 8 features:

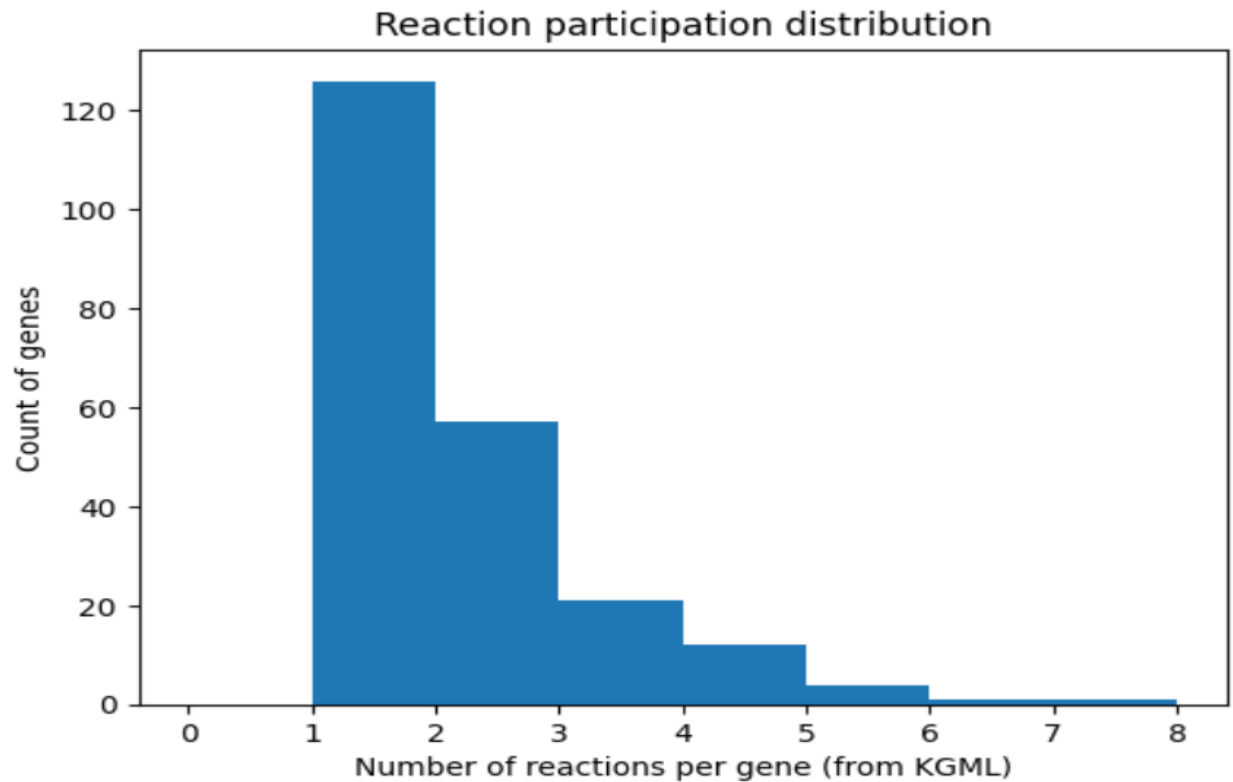
| Feature     | Description                                         |
|-------------|-----------------------------------------------------|
| col00       | log1p degree (graph connectivity)                   |
| col01–col06 | Pathway membership binary indicators for 6 pathways |
| col07       | Bias constant = 1                                   |

All features (except bias) were normalized to mean=0, std=1. Correlation analysis with labels shows:

- pathway\_01200 positively correlated (0.348)
- pathway\_00620 negatively correlated (-0.204)
- Bias constant shows no correlation

## 2.4 Label Distribution

| Label | Count | Ratio |
|-------|-------|-------|
| 0     | 126   | 0.568 |
| 1     | 96    | 0.432 |



## 2.5 Train/Validation Split

- Training: 178 nodes (101 class0, 77 class1)
- Validation: 44 nodes (25 class0, 19 class1)
- Validation ratio: 0.2
- Random seed: 42

No overlap between train and validation sets.

## 3. Model Implementation

### Architecture

- Model Type: Graph Convolutional Network (GCN)
- Layers: 2 hidden layers, 64 hidden dimensions each
- Dropout: 0.35
- Activation: ReLU
- Output: Binary classification (Label0 vs Label1)

The GCN aggregates neighboring node information and combines it with node features to learn embeddings for prediction.

Message Passing Formulation:

$$H^{(l+1)} = \sigma(\widehat{D}^{-1/2} \widehat{A} \widehat{D}^{-1/2} H^{(l)} W^{(l)})$$

where  $\widehat{A}$  is adjacency with self-loops,  $\widehat{D}$  is degree matrix, and  $W^{(l)}$  is learnable weight.

Device

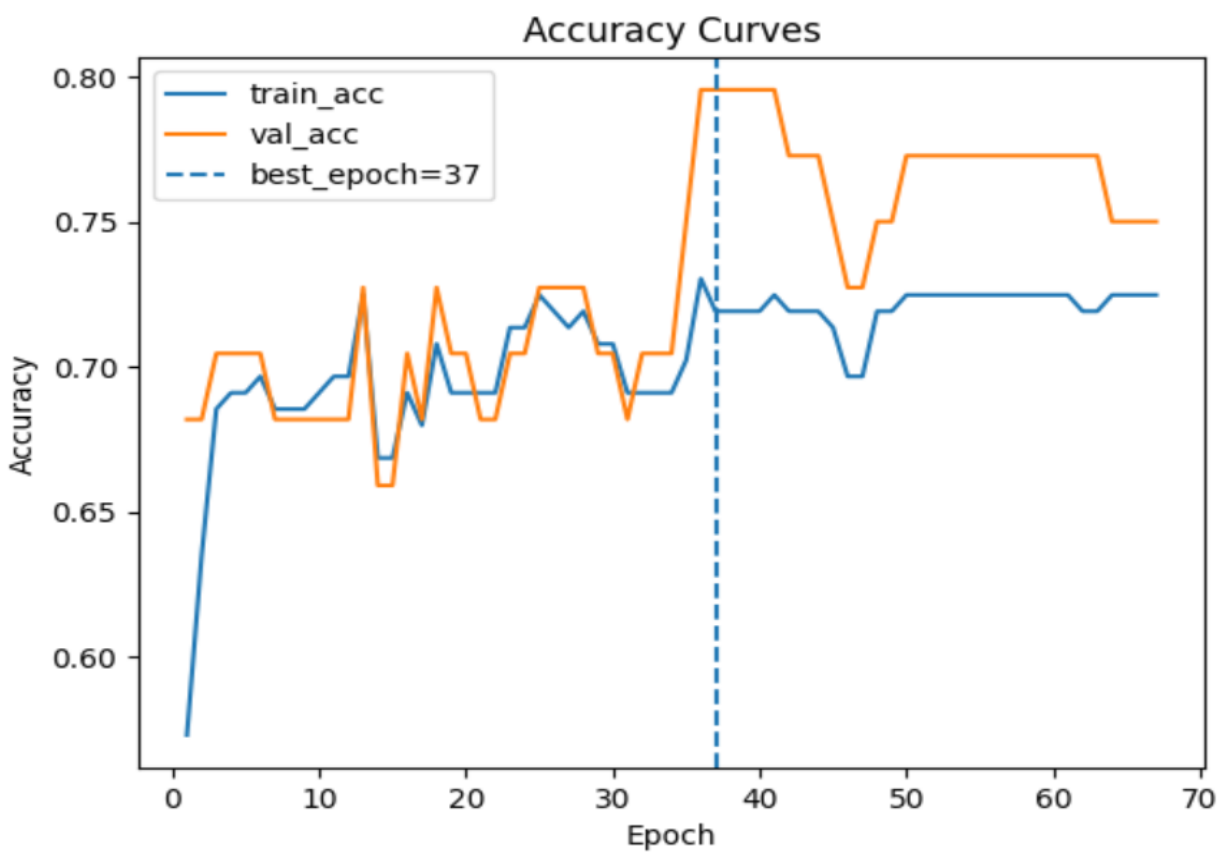
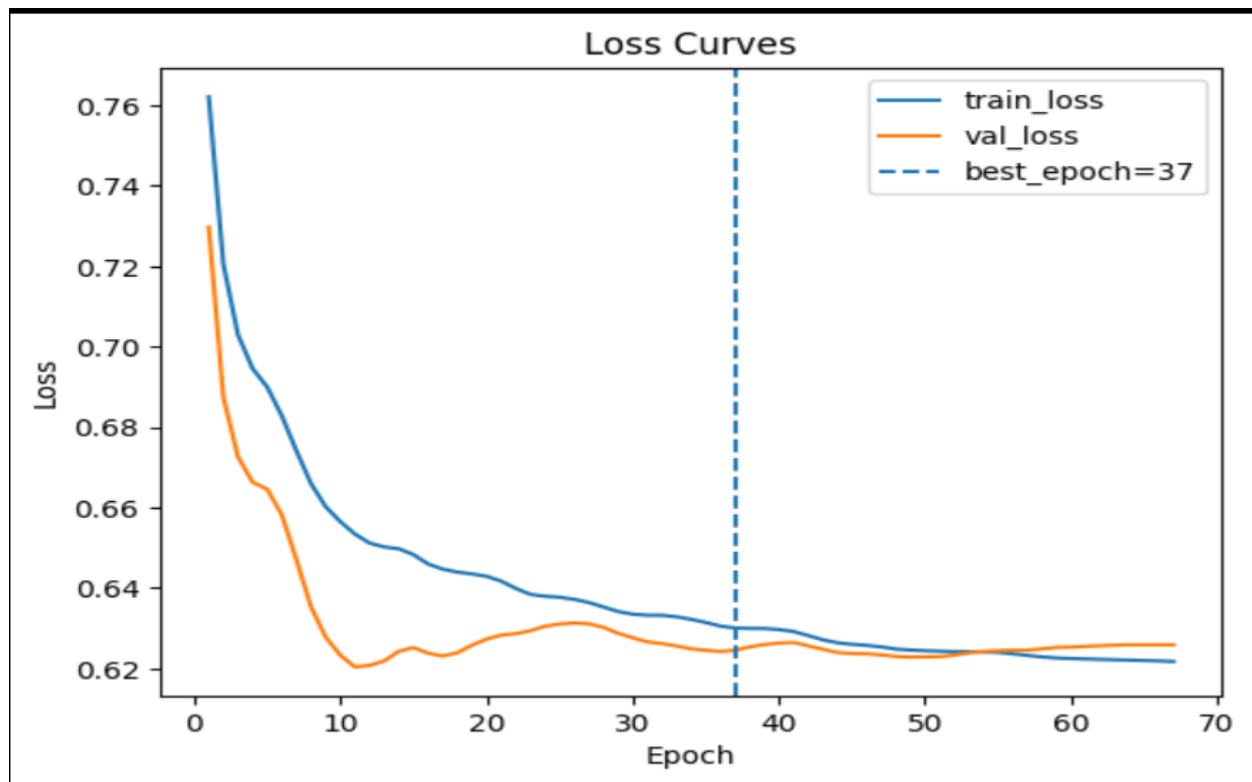
Training was performed on CPU as no GPU was detected.

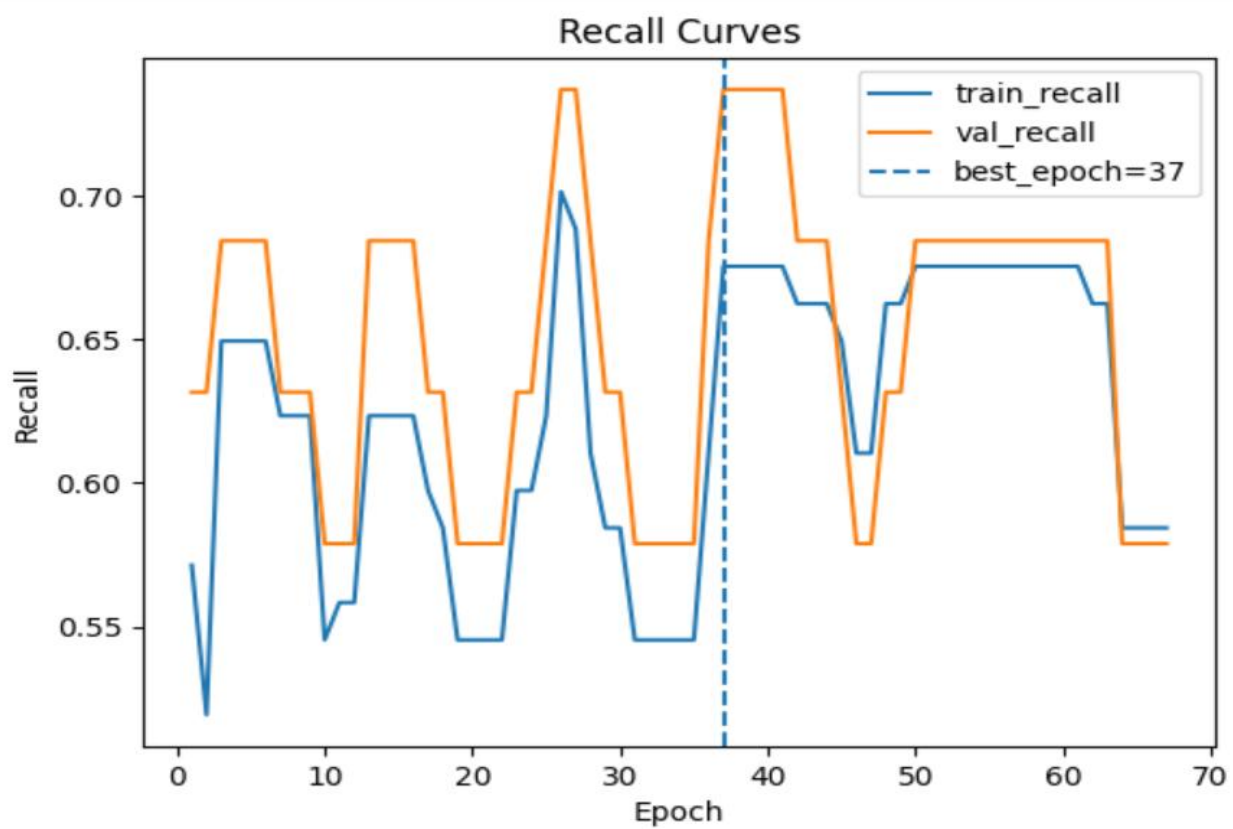
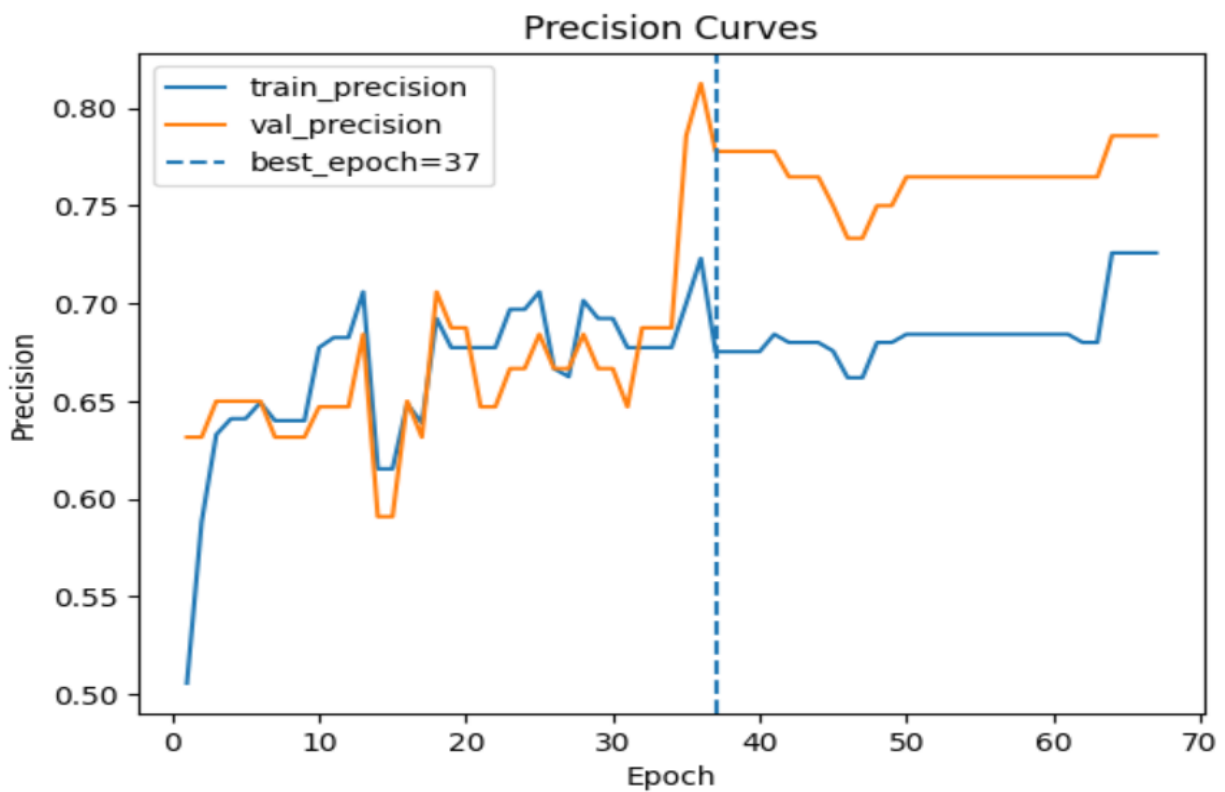
#### 4. Training Configuration and Hyperparameter Tuning

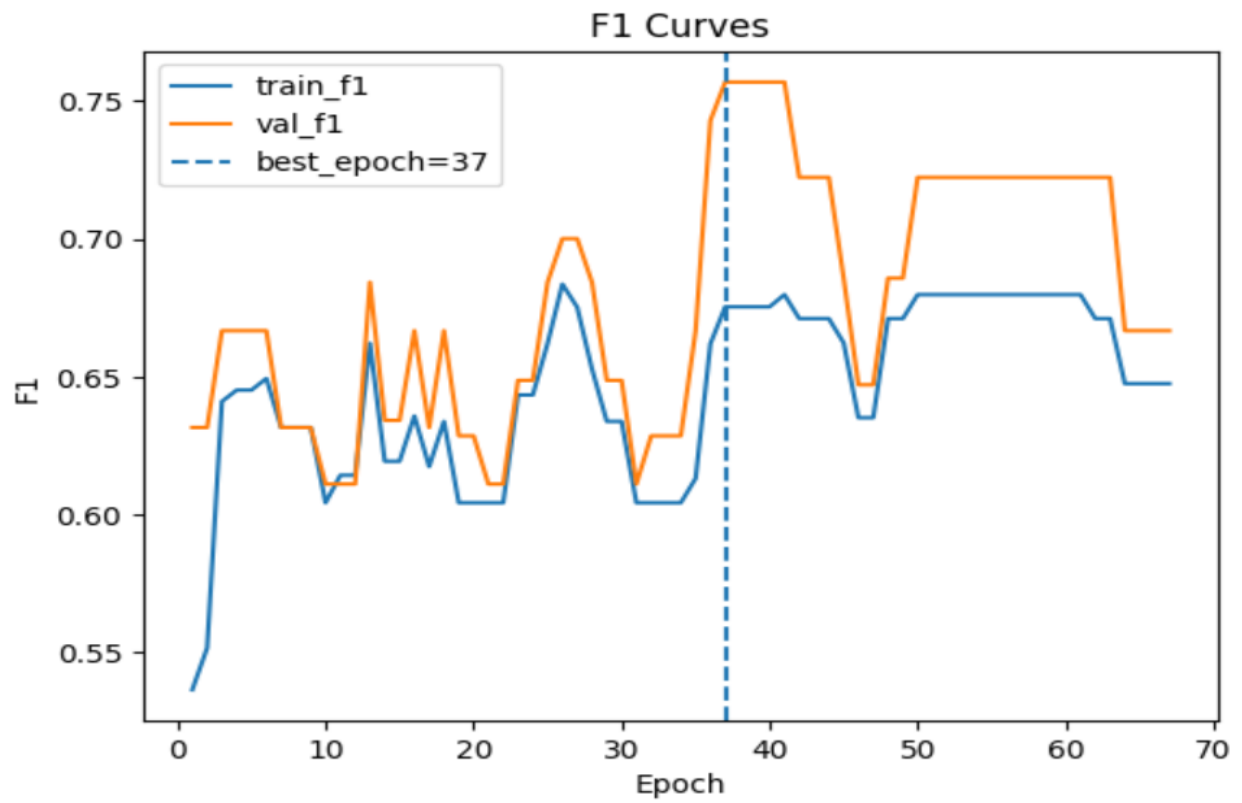
| Hyperparameter    | Value                           |
|-------------------|---------------------------------|
| Optimizer         | Adam                            |
| Learning Rate     | 0.01 (step decay applied)       |
| Weight Decay      | 0.0005                          |
| Gradient Clipping | 1.0                             |
| Epochs            | 250                             |
| Early Stopping    | Patience = 30, monitor = val F1 |
| Loss Function     | Binary Cross-Entropy            |
| AMP               | Disabled                        |

Training Observations:

- Best epoch: 37 (validation F1 = 0.757)
- Early stopping triggered at epoch 67







## 5. Experimental Results

### 5.1 Training Performance

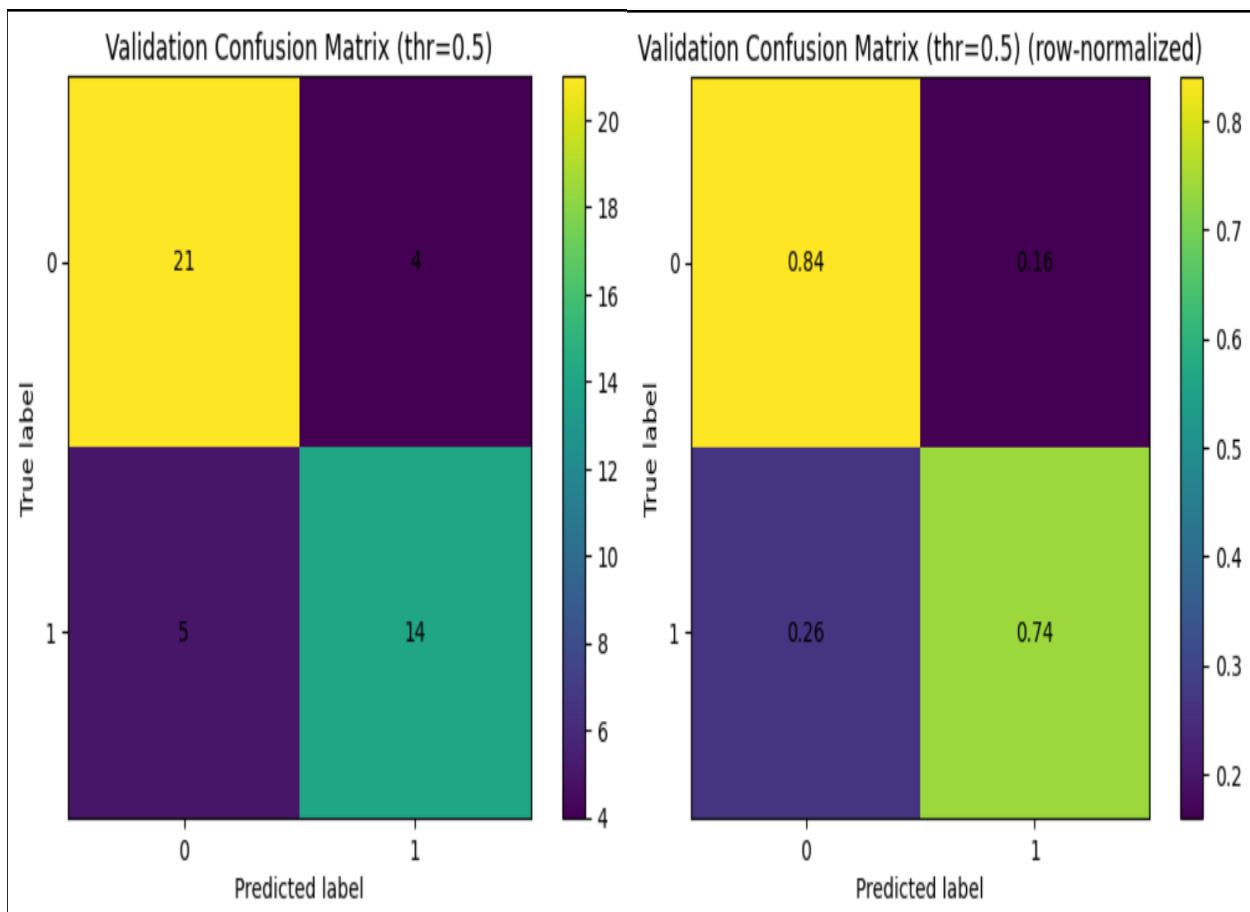
| Metric            | Value |
|-------------------|-------|
| Accuracy          | 0.719 |
| Precision         | 0.675 |
| Recall            | 0.675 |
| F1 Score          | 0.675 |
| Balanced Accuracy | 0.714 |
| AUROC             | 0.779 |

## 5.2 Validation Performance (Best Epoch 37)

| Metric            | Value |
|-------------------|-------|
| Accuracy          | 0.795 |
| Precision         | 0.778 |
| Recall            | 0.737 |
| F1 Score          | 0.757 |
| Balanced Accuracy | 0.788 |
| AUROC             | 0.741 |
| AUPRC             | 0.795 |

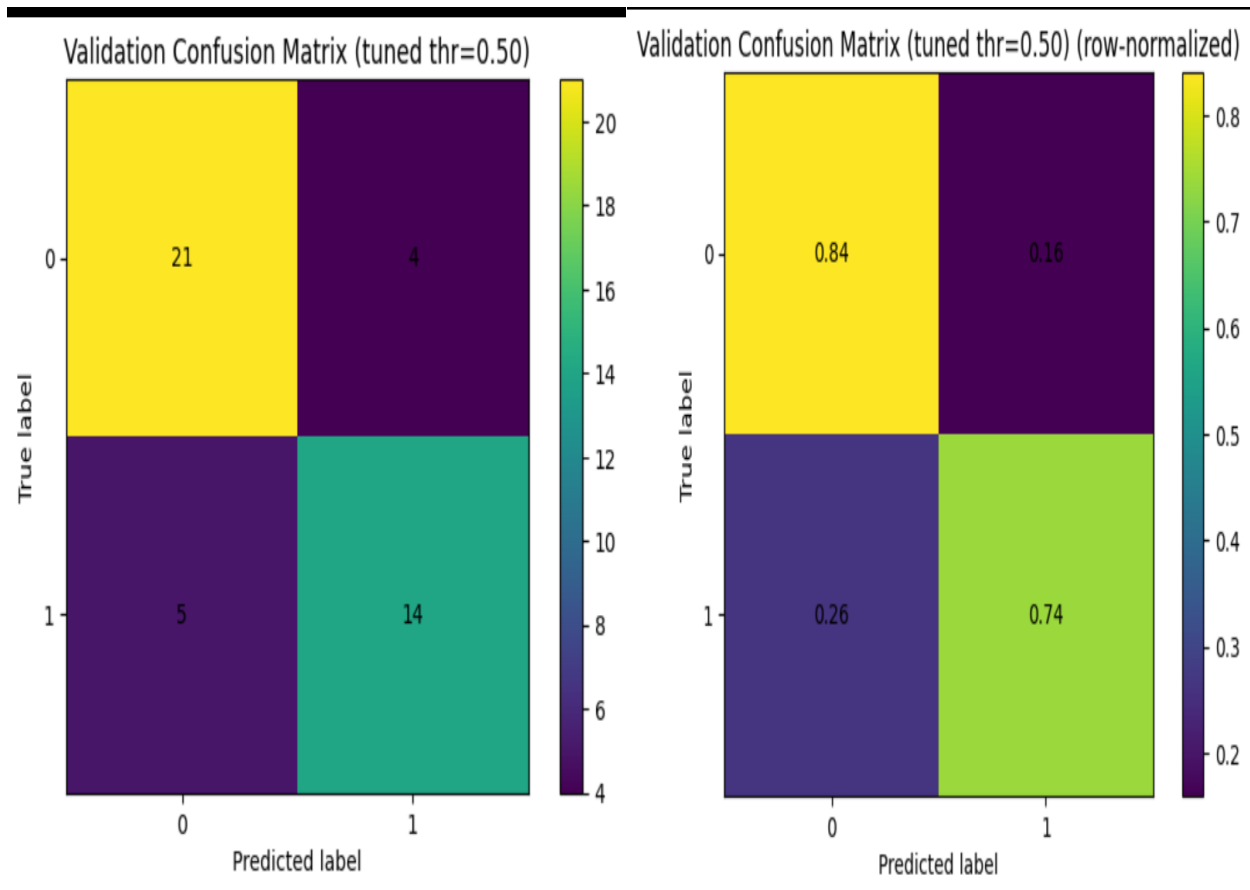
Validation Confusion Matrix (Threshold = 0.5)

|        | Pred 0 | Pred 1 |
|--------|--------|--------|
| True 0 | 21     | 4      |
| True 1 | 5      | 14     |



## 5.3 Threshold Tuning

- Best F1 achieved at threshold = 0.5
- No further improvement obtained with alternative thresholds



## 6. Discussion and Analysis

### 6.1 Graph Diagnostics

- Sparse graph (edge density 0.024) limits propagation of information
- 98 isolated nodes rely solely on intrinsic features, reducing predictive power
- Largest component contains 124 nodes (~56% of graph)
- Degree distribution: min=0, median=4, mean≈5.32, max=19

## 6.2 Feature and Label Insights

- Degree and pathway membership features contributed most to model predictions
- Bias constant was uninformative
- Moderate correlation between specific pathways and labels validates biological relevance

## 6.3 Model Performance

- GCN effectively learned from connected nodes; validation F1=0.757 demonstrates good predictive capability
- Training and validation metrics indicate minimal overfitting
- Early stopping and dropout helped stabilize training

## 6.4 Limitations

- CPU-based training is slower than GPU
- Sparse and disconnected graph limits neighborhood aggregation
- Only six pathways were used; adding more pathways could improve signal

## 7. Conclusion

- Implemented a GCN from scratch to classify enzyme reaction participation
- Achieved balanced F1 = 0.757 and AUROC = 0.741 on validation set
- Graph structure and pathway-based features significantly influenced performance
- Early stopping and dropout prevented overfitting, even on a small dataset

This study demonstrates that GNNs are suitable for metabolic pathway analysis and highlights the importance of graph connectivity and feature selection in biological network prediction.